

Unit 10, Lesson 6: Outliers

Benchmark

MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.

Learning Targets

- ☐ I can determine how data is impacted by an outlier.
- ☐ I can relate the impact of an outlier to which measures of center and variation best describe a distribution.

10.6.1 Warm-Up: You Tell the Story

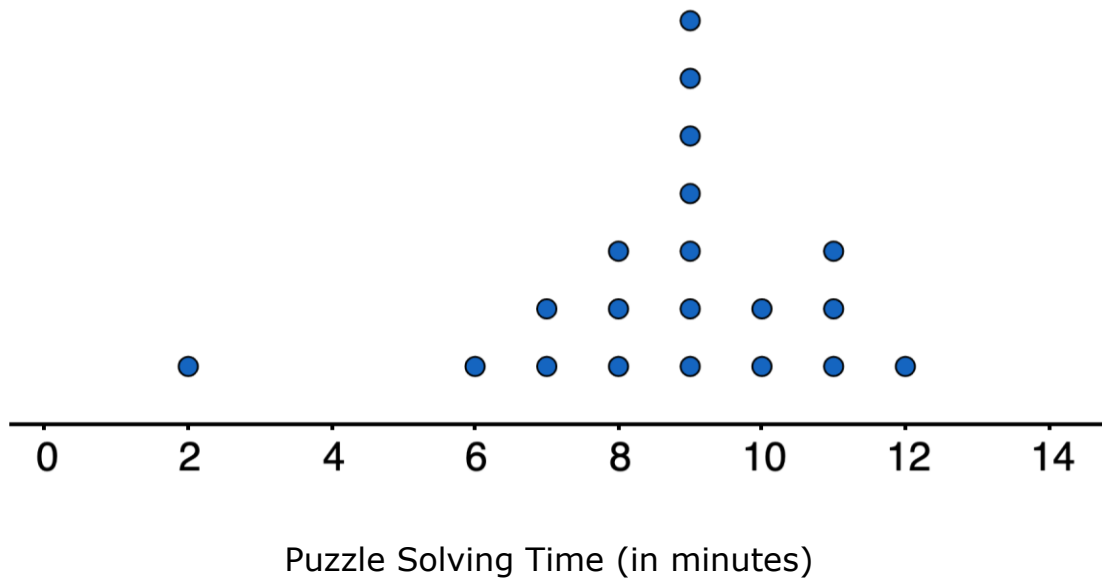
1. Robbie goes into a restaurant and orders a deli sandwich and a cola for lunch. Afterward, he pays the bill, tips his waitress, and goes outside. He slowly takes in his surroundings. The sky is black and the city streets are deserted. What happened?

10.6.2 Exploration: Data Values

Work with your partner to complete the following.

1. Twenty students timed how long it took each of them to solve a puzzle. Below is a list of their solution times, in minutes, and a dot plot displaying the data.

2, 6, 7, 7, 8, 8, 8, 9, 9, 9, 9, 9, 9, 9, 10, 10, 11, 11, 11, 12



- a. Describe the distribution of data.
- b. Determine the mean of the data set. Then, draw an arrow ($\uparrow$) below the number line that points to this value.
- c. Determine the median of the data set. Then, circle the value on the number line that represents this value.

- d. Discuss with your partner what you notice about the mean and the median of the data. Summarize your discussion.
- e. What is the range of the data?
- f. Find the interquartile range (IQR).

Q1	Q3	IQR

2. Imagine that the fastest time of 2 minutes was removed from the data set. Discuss with your partner what effect, if any, removing this value has on the values of center and variation (mean, median, range, IQR). Summarize your discussion.
3. How does the shape of the graph change after 2 is removed?

4. Compare the new values with the original values.

	Original Value	$>, <, =$	New Value after 2 is Removed
Mean			
Median			
Range			
IQR			

10.6.3 Bring It Together: Outlier Effects

1. Complete each statement to describe the effect the outlier, 2, has on each measure of center or variation.

- The presence of an outlier does not have a significant effect on the ☐ mean.
☐ median.
- The presence of an outlier will change, often significantly, the value of the ☐ mean.
☐ median.
- The presence of an outlier does not have a significant effect on the ☐ IQR.
☐ range.
- The presence of an outlier will change, often significantly, the value of the ☐ IQR.
☐ range.

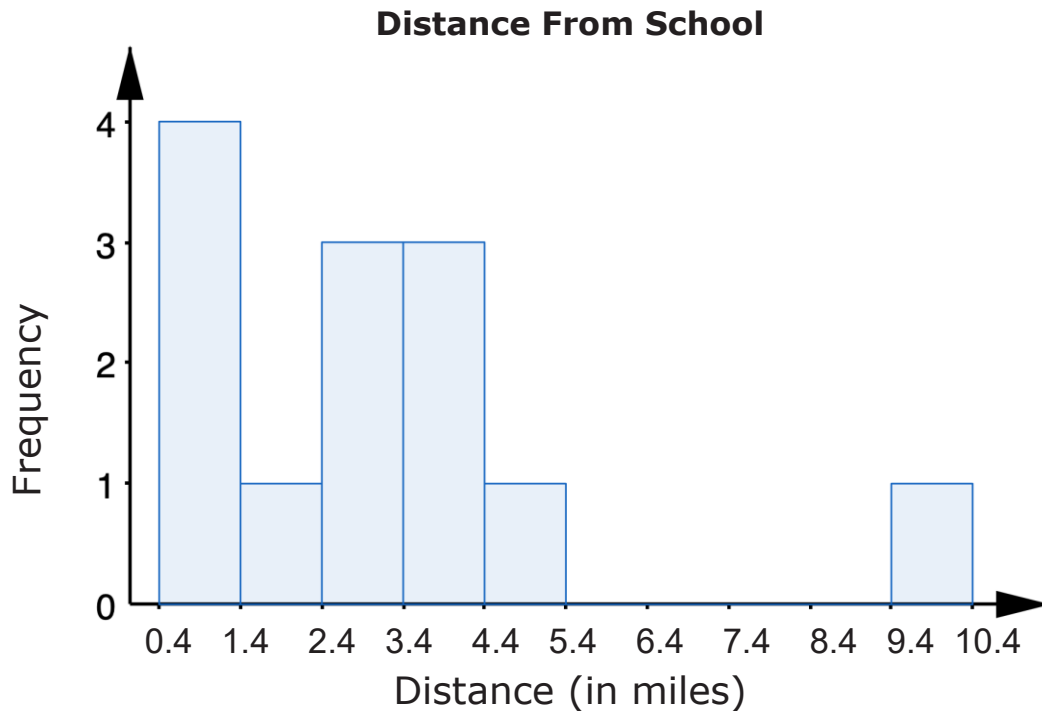


When outliers are present or a distribution is skewed, the _____ is the preferred measure of center and the _____ is the preferred measure of variation to describe the distribution.

10.6.4 Guided Practice: Data Outliers

1. Zane and 12 of his friends listed how far, in miles, they live from the school and created a histogram.

0.4, 0.5, 1, 1, 2.4, 2.7, 3, 3.2, 4, 4.2, 4.4, 5, 9.5



- a. What is the height of the bar for the interval 9.4 – 10.4?
- b. Find the mean and the median distance from school.
- c. Discuss with your partner what effect that outlier has on the mean, median, range, and interquartile range of the data. Summarize your conclusions.

- d. Noelle is Zane's friend who lived 9.5 miles away from the school. She recently moved closer, and now only lives 5.5 miles away from the school.

The data set with Noelle's new distance from school, in purple, is shown.

0.4, 0.5, 1, 1, 2.4, 2.7, 3, 3.2, 4, 4.2, 4.4, 5, 5.5

Would Noelle's new distance from school have similar effects on the data? Explain your reasoning.

- e. What measure of center best describes the typical distance Zane's friends live from school?
- f. What measure of variation best describes the spread of the data distribution?

10.6.5 Your Turn!

1. Zoe needed exact change to purchase a snack and a drink from the vending machine. She asked her friends how much change, in dollars, they had and listed the results.

0.15, 0.15, 0.20, 0.25, 0.25, 0.30, 0.30, 0.30, 0.35, 0.75, 0.85

- a. What do you notice about the data?
- b. Will the mean be higher, lower, or approximately the same as the median? Explain your reasoning or justify your answer by showing your work.
- c. Describe the impact of removing 0.85 from the data set.
- d. The values 0.85 and 0.75 are outliers in this data set. What happens to the range if 0.85 and 0.75 were both removed from the data set?
- e. The IQR of the full data set is _____.
The IQR with 0.85 and 0.75 removed is _____.
- f. Is there a big difference between the IQR of the original data set and the modified data set?

10.6.6 Wrap-Up: Cool Down

1. The presence of an outlier has a significant effect on which measures of center and spread? Select all that apply.
 - ☐ Mean
 - ☐ Median
 - ☐ Interquartile Range
 - ☐ Range

2. A data set has an outlier that is significantly less than most of the other values.
 - a. Explain how this impacts the measures selected in number 1.

 - b. Which measures of center and variation would you use to best describe the data?